EXAMINATION

Luleå University of Technology

Course: D7024E
Course name: Distributed and Mobile Computing Systems
Date: 2024-10-30
Time: 13.00-17.00, online home exam
Aid: WWW and Generative AI (e.g., ChatGPT, Bard) (no interactive communication with other people)

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Grade scale: 50% grade 3, 75% grade 4 and 90% grade 5	Note that this grade is for the exam. The exam is supposed to be easy enough so that anyone that followed the course properly should pass it. The grade for the course (including all assessment criteria) may be different from the grade of the exam.
Total number of questions and score: 2 questions, total score 36	
Other information: This is an individual home exam.	Note: No interactive cooperation with other people is allowed. Information can be searched on the Web and with generative AI (e.g., ChatGPT, Bard, etc.). Please provide references when such information is used in your answers.

General instructions

You shall formulate answers with your own words; if you use generative AI a personal reflection must be included. In any case reference your primary sources (e.g., URL). Your **solution must be uploaded in the Canvas D7024E room under Assignments**, at the latest at 17:00 (unless you have gotten approval for extra time). **Please put your name both in the filename and on the first line of your document.**

After examination

Examination results are posted on Canvas within 4 weeks after the examination.

1. **Mixed questions on distributed systems (10p)**

   These are multi-choice questions, with three alternatives. Please answer "true", "false" or “depends” to each of the following statements. If your answer is “depends” you must explain briefly to get credits. The answer “true” or “false” requires no explanation unless you want to explain some assumed conditions. If you have identified a corner case and provide a good explanation, we can accept an answer that is different from the expected one. Each question is worth 0.5p

a. Marshaling (a k a serialization) is used to get global consensus on the ordering of events.

b. In distributed databases, a transaction log (or command log) is needed to handle consensus.

c. In databases, deadlocks can be resolved by time-outs and rolling back performed commands.

d. NOSQL data stores typically trade off client-defined transactions for lock-free operations.

e. Byzantine global consensus solutions typically handle a majority of traitors.

f. To obtain consensus in a distributed system you must make assumptions on the connectivity, and the fraction of nodes being alive.

g. NAT/firewall traversal by TURN can sustain peer-to-peer communication between two peers behind symmetric firewalls.

h. A DNS provider may handle heartbeats from alternative webservers for a given ULR to automatically drop unhealthy ones from its DNS responses.

i. Providing immutable messages is a key paradigm in distributed systems, where messages can be reliable or unreliable.

j. Leader-based architectures including automatic leader elections are often used for scalability where immediate or fast data consistency is required.

k. Rendezvous handshaking is a typical approach in asynchronous systems.

l. NOSQL databases typically provide better data locality than relational databases.

m. Message queues and pub-sub provide the same message semantics, both based on subscriptions.

n. Scalable reliable multicast (SRM) is based on negative acknowledgments.

o. For scaling out a system with eventual consistency, the performance is improved if we adopt a notion of transactions to support multiple commands atomically.

p. PGP uses only asymmetric keys in normal operation.

q. In ad-hoc mobile systems, the nodes using proactive routing typically discover an end-to-end path before they transmit a packet.

r. Data storage that is based on consistent hashing places data so that short round trip times are obtained.

s. Map-reduce in Hadoop is designed to move computation to where data is stored, rather than the other way around.

t. Stream analysis (e.g., Apache Spark) and map-reduce (e.g., Apache Hadoop) provide similar properties in most cases.

2. Message systems, microservices and blockchain (26p)

a) Related to mutability and immutability, what are the three overall abstractions that most distributed systems build on? (6p)

b) In global consensus, explain the command log abstraction, what essential properties it provides and how these properties are ensured. Elaborate on the differences between Bysantine and Non-Bysantine (6p)

c) How do massively online multiplayer games scale their game to actually be massive? What are some important trade-offs due to scaling? Is it possible to scale indefinitely? If yes, how? If not, why not? (6p)

d) Ethereum uses a Proof-of-Stake (PoS) consensus mechanism, which relies on validators instead of miners. Compare the performance, scalability, and security aspects of PoS with traditional Proof-of-Work (PoW) in the context of Ethereum. There are many aspects/dimensions for performance, scalability, and security so please be clear about which aspects you refer to. (4p)

e) When increasing the number of nodes and message overhead, analyze the trade-offs between performance, scalability, and security in PBFT. There are many aspects/dimensions for performance, scalability, and security so please be clear about which aspects you refer to. (4p)

Best Wishes!
Finally, please fill out the **online course evaluation in Canvas!**